



Bangladesh Mathematics and Science Olympiad

BdMSO Rules and Regulations

May 10, 2026

Organizer

Society for the Popularization of Science, Bangladesh (SPSB)
Bangladesh Open Source Network (BdOSN)

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1. Introduction

Bangladesh Mathematics and Science Olympiad (BdMSO) is designed to introduce primary school students to the world of mathematics and science from an early age, nurturing their natural curiosity and problem-solving skills. BdMSO serves as the official qualifying platform to select primary school students (up to Grade six) to represent Bangladesh internationally. Each year, the competition selects a national team of twelve students, six for Mathematics and six for Science, to participate in the International Mathematics and Science Olympiad (IMSO). In 2026, the 23rd IMSO will be held in Indonesia.

2. General Provisions

The following general provisions outline the operational standards, examination protocols, and logistical requirements governing all stages of the competition.

2.1. Official Language

English is the official language for all examination papers, instructions, and official communications throughout all phases of the Olympiad.

2.2. Exam Structure

The National Olympiad is a two-hour single exam based on the BdMSO Syllabus, covering both Mathematics and Science sections. The Team Selection Test (TST) follows the IMSO exam standard procedure and involves a day-long evaluation. Separate TSTs for Math and Science are held on different days. Detailed information is available in the [Question Pattern](#) section of this document.

2.3. Academic Ethics

All examination papers must be the original, independent work of the participant. The use of calculators, mobile phones, smartwatches, or any other electronic communication devices is strictly prohibited. Any breach of academic integrity will result in immediate disqualification. Participants must strictly adhere to all instructions provided by invigilators on the ground.

2.4. Document Integrity

All information provided during registration must be accurate. Any discrepancy discovered regarding academic status or age during any phase of the competition will lead to immediate disqualification without a refund of fees. The organizers reserve the right to request official school transcripts, birth registration certificates, or identity documents at any stage.

2.5. Syllabus Authority

Problems are constructed based on the [BdMSO Syllabus](#). While they align with the National/British curricula, they incorporate advanced intellectual reasoning and creative problem-solving concepts that extend beyond standard textbooks.

2.6. Evaluation

Answer scripts are evaluated by a centralized academic team. Scoring and ranking decisions are final and are not subject to appeal or re-evaluation. However, participants may contact the organizers regarding technical discrepancies in personal data (e.g., spelling of name or school) within 48 hours of result publication.

2.7. Results and Winners

National Round winners will be awarded on the same day as the competition, and the finalized list of winners will be uploaded on the official website within one week. Awards will be granted to approximately the top 20-30% of registered participants. Individual rankings, raw scores and results for participants who do not secure a medal will not be published publicly.

2.8. Documentation Readiness

As the final team represents Bangladesh internationally, all participants attending the TST and interested accompanying parents are strictly advised to have their travel documents ready. Any mismatch in documentation or failure to provide requisite papers in a timely manner may cause immediate disqualification from the delegation. Required documents typically include a valid E-Passport or Machine-Readable Passport (MRP), updated bank statements for visa financial requirements, a guardian affidavit attested by a Notary Public, official school certification or ID, and a No Objection Certificate (NOC) from the employer for accompanying parents. The organizers are not liable for disqualifications arising from missing or inconsistent documentation.

2.9. Health and Safety

As the competition includes residential camps and an international round, the safety of participants is of utmost priority. During these phases, students and guardians will be accommodated in separate zones or facilities. Guardians hold the sole responsibility to disclose any pre-existing medical conditions or allergies of the participant at the time of registration. While organizers will provide basic first aid support, they are not liable for major medical emergencies or pre-existing health issues.

2.10. Sponsorship and Branding

The organizers reserve the right to engage sponsors and partners for the competition. Participants and guardians agree to comply with branding and activation requirements as outlined in Memorandums of Understanding (MoUs) or partnership agreements. This may include participating in sponsored photo sessions, wearing partner-branded apparel, or attending designated sponsor events.

2.11. Conflict of Interest

No individual may serve on the Team, as a Mentor, or as an Invigilator if they have a first-blood relative participating in the competition. Furthermore, Volunteers involved in the competitions have no influence over the final Team Selection Test (TST) scoring or the final selection process of the National Team.

2.12. **Media and Data Privacy**

By registering for the competition, guardians formally acknowledge and consent to the organizers' use of the participant's name, school, and contact details for official updates and further communication. Furthermore, the organizers reserve the right to use photographs and video recordings of participants for promotional purposes across official websites, social media, and press releases.

2.13. **Financial Liability**

All registration fees for the National Round, Camp, and TST are strictly non-refundable. The organizers are not responsible for personal travel expenses, loss of personal property, or external costs incurred by participants or guardians.

2.14. **Right to Amend**

The organizers reserve the right to amend these regulations, competition schedules, or venue arrangements at any time to ensure the smooth operation of the Olympiad. Any such changes will be communicated via the official website and shall be binding upon all participants.

3. **Participation Criteria**

BdMSO is open to students across Bangladesh who meet specific age and grade requirements. Eligibility is strictly governed by the cutoff date of October 1, 2026.

3.1. **Eligibility Requirements**

- Participants must be 13 years old or younger.
- Participants must be born on or after October 1, 2013.
- Participants must be enrolled in grade 6 or lower as of October 1, 2026.

Note: For students in the British Curriculum (or similar international streams) who are promoted to Grade 7 in July/August 2026, they are **not eligible** to participate, even if they were in Grade 6 during the registration period in May.

3.2. **Legacy Provisions:** The organizers reserve the right to invite previous IMSO Bangladesh Team members to participate in the selection process starting directly from the National Residential Camp, regardless of their current National Round participation or results, provided they remain eligible by age and grade.

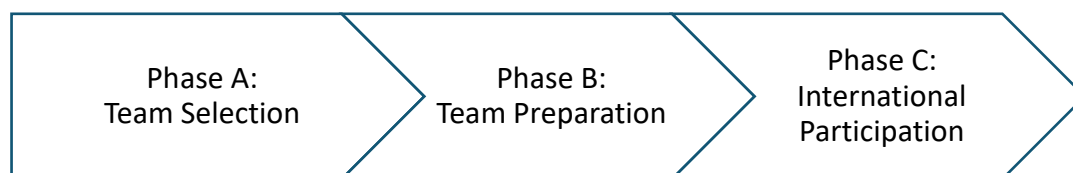
3.3. **Categorization:** BdMSO operates as a single category for IMSO qualification. To encourage foundational logic and early interest in STEM, the organizers may introduce a Quiz Competition for students in Grade 3 or lower. This is designed purely for encouragement and functions solely as a national quiz competition in STEM. Performance in the Quiz does not lead to the National Camp or Team Selection Test, unless a gifted student explicitly opts to compete in the standard Olympiad and meets the required benchmarks.

4. Competition Timeline

The BdMSO cycle will roughly follow this year-round schedule:

Month	Segment
January – February – March	Foundation Courses
April – May	Announcement, Activation, Preparatory Workshop, Registration
June	National Olympiad
July	National Residential Camp, Team Selection Test (TST)
August – September	IMSO Team Preparatory Camp
October	International Participation (IMSO)
November – December	Foundation Courses

5. Competition Structure



The BdMSO cycle is divided into three major phases: Team Selection, Team Preparation, and International Participation.

Phase A: Team Selection

The Team Selection process is a formal, multi-stage progression designed to identify the most promising young talents. It begins with individual online registration, followed by the National Qualifying Round and the National Residential Camp, concluding with the Team Selection Test (TST) to determine the final international delegation.

I. National Olympiad (Qualifying Round)

Prospective participants must complete online registration via the [official portal](#). The National Olympiad will be held in multiple regional venues, either simultaneously or on different dates. The assessment is a two-hour written exam combining both Mathematics and Science sections, featuring a mix of Multiple-Choice Questions (MCQ), Short, and Broad Questions. (See [National Round Question Pattern](#)).

This competition day typically commences with the National Anthem, followed by the exam, a lunch break, a cultural program, and a prize giving ceremony. Invitation letters for the National Residential Camp are handed over to the guardians of medalists immediately following the ceremony on the same day.

II. National Residential Camp

All winners from the National Olympiad are invited to a three-day National Residential Camp. This activity-based learning camp focuses on advanced problem-solving and provides generic training for all attendees with subject separation to prepare students for the TST.

A participation fee is required for students to attend this camp. A maximum of one guardian per student is permitted to attend the camp by paying additional fee. During the camp, students and guardians stay in separate zones with limited meeting times. Guardians must attend dedicated sessions regarding camp norms and the detailed team selection process, including cost structures, previous IMISO experiences and international participation rules.

III. Team Selection Test (TST)

TST held approximately seven days after the conclusion of the National Camp. It is the final and exclusive examination to determine the National Team. TST adheres strictly to the BdMSO Syllabus, and IMISO standards. The TST consists of separate day-long evaluations for Mathematics and Science conducted on two different days. Participants may sit for the Mathematics TST, the Science TST, or both. Prospective examinees are required to pay a TST Examination Fee of 1,000 BDT per subject.

Selection Finality: The organizers do not publish individual TST scores publicly. A list identifying the top 15 participants in Mathematics and the top 15 participants in Science is announced. The primary National Team is finalized by selecting the top six students in each subject based on their TST, National Round and Camp performance. Vacancies are filled chronologically by the next eligible member from the merit list.

Expanded Delegations: Subject to seat availability and IMISO Host Country regulations, the organizers reserve the right to invite additional members from the merit list to expand the team (e.g., 9+9, 10+10, or 12+12). While the merit list chronology serves as the primary basis for these invitations, the organizers maintain the final authority to evaluate candidate suitability to ensure the delegation's overall composition meets international standards. These additional participants generally incur significantly higher international registration fees than the primary 6+6 members.

Phase B: Team Preparation

Once finalized, the National Team begins phase two with a formal orientation meeting for selected members and guardians to address administrative requirements (payments, deadlines, passports and academic records). Following orientation, the team enters a mandatory two-month intensive training period involving regular online and offline classes.

A four-day residential camp is organized specifically for team building, IMISO-style problem solving, and in-depth preparation. The focus of this training remains on advanced topics, IMISO-style problem solving, and in-depth preparation. Attendance and active participation in all sessions are mandatory to maintain team status.

Phase C: International Participation

The team represents Bangladesh at the 23rd International Mathematics and Science Olympiad (IMSO) to be held in Indonesia. The total duration of the international participation is usually around seven days. All delegation members, including students and accompanying parents, must strictly abide by the instructions of the Team Leaders and Tutors throughout the entire trip.

The organizing authority will nominate a standard delegation to the host country consisting of 15 members: 12 Student Participants (6 Math, 6 Science), 2 Tutors (1 Math, 1 Science) & 1 Team Leader. The organizers are authorized to extend team members including additional Mathematics or Science participants, Team Leaders, Mentors, and Accompanying Parents.

6. Syllabus, Marks Distribution & Question Pattern

To ensure consistency and prepare participants for international standards, the BdMSO problems are constructed based on international benchmarks. While they align with the National and British curricula, they include advanced intellectual reasoning and creative problem-solving that extend beyond standard textbooks. The official language for the contest is English.

- The official BdMSO Syllabus can be found in [ANNEX I](#) and [ANNEX II](#).

National Round

The BdMSO National Round is an individual competition. The assessment consists of a single two-hour written exam combining both Mathematics and Science sections.

- The specific question-type sample questions can be found in [ANNEX III](#) and [ANNEX IV](#)

Marks Distribution: BdMSO National Round			
Duration: 120 Minutes			
Section	Question Type	Distribution	Marks
Mathematics	Short Answer Problems	20 Questions	20
	Essay Problems	10 Questions	20
	Exploration Problem	2 Questions	10
	1. Marks: Mathematics		50
Science	Multiple Choice (MCQ)	20 Questions	20
	Short Answer Problems	10 Questions	20
	Brief Problems	2 Questions	10
	2. Marks: Science		50
3. Total Marks (1+2): Mathematics and Science			100

Team Selection Test

The Team Selection Test follows the IMSO international evaluation standard. The TST is a day-long event consisting of three separate exams per subject. Snacks and refreshment breaks will be provided between each session.

- **Mathematics TST:** Consists of three parts: Short Answer, Essay Type, and Exploration problems.
- **Science TST:** Consists of three parts: Theory 1, Theory 2, and Experimental problems.
- For sample questions regarding the TST, please visit **Annex IV**.

Marks Distribution: Team Selection Test				
Subject	Section	Distribution	Duration	Marks
Mathematics	Short Answer Problems	25 Questions	60 Minutes	25
	Essay Problems	13 Questions	90 Minutes	39
	Exploration Problems	6 Questions	120 Minutes	36
	Total Marks: Mathematics			100
Science	Theory 1 (MCQ)	30 Questions	60 Minutes	30
	Theory 2 (Open Ended)	9 Questions	90 Minutes	45
	Experiment (Lab)	2 Questions	120 Minutes	25
	Total Marks: Science			100

7. Evaluation and Selection Process

The National Round serves as the primary screening mechanism. Selection is conducted through a structured statistical evaluation process. Every question is evaluated based on a standardized marking scheme. For multiple-choice and short-answer questions, only the final submitted answer is considered. For descriptive/essay-type problems, partial credit may be awarded for valid logical deductions and procedural steps.

Award Classification

Participants achieving qualifying scores are designated as "Winners". To ensure the elite recognition and subject-specific talent identification, winners are classified based on their sectional performance:

- **All-Round Winners:** Participants who rank within the top 10% of the Mathematics section and the top 10% of the Science section independently.
- **Mathematics Winners:** Participants who rank within the top 10% of the Mathematics section but do not qualify as All-Round winners.
- **Science Winners:** Participants who rank within the top 10% of the Science section but do not qualify as All-Round winners.

Elevation Rule: The organizing team reserves the right to top up the winner brackets to the next highest-ranked student if the 10% threshold does not meet the minimum required headcount for the National Camp.

Tie-Breaking Protocol

In the event of identical scores among participants, the organizing team shall determine the final ranking based on the following sequential criteria:

- **Score Balance:** Preference is awarded to the participant demonstrating a more balanced score distribution between the Mathematics and Science sections.
- **Difficulty Index:** Priority is granted to the participant who successfully solved problems designated with a higher difficulty coefficient on the exam.
- **Age:** The chronologically younger participant Should, determined by exact date of birth, shall be awarded the higher rank.

Special Trophies

Champion Trophies are awarded to the absolute highest-scoring individuals in each respective section. These are awarded on the final day of the National Residential Camp. Individual participant is strictly limited to receiving a maximum of one trophy. In instances where a participant achieves the highest score across multiple categories, the All-Rounder Trophy supersedes other awards and is granted as the highest overarching honor.

Camp and TST Participation

National Round Winners shall receive invitations to the National Residential Camp and the subsequent Team Selection Test (TST).

- **Mandatory Subject Alignment:** Invited participants are obligated to attend academic sessions and undertake the TST in the specific discipline (Mathematics or Science) corresponding to their highest score during the National Round. For All-Round Winners, the mandatory assigned subject is dictated by their higher sectional score.
- **Optional Subject Participation:** Participants may elect to attend sessions or sit for the TST in their secondary subject as an auxiliary activity. However, strict adherence to the schedule and requirements of their primary assigned subject remains mandatory.

National Team

The primary National Team is finalized by selecting the top six students in each subject based on a weighted average of their National Round, Camp performance, and TST. Subject the IMSO Host Country regulations, organizers reserve the right to invite additional members from the merit list to expand the team. The organizers maintain the final authority to evaluate candidate suitability. These additional participants generally incur significantly higher international registration fees than the primary 6+6 members.

8. Awards and Recognition

National Round: Approximately the top 20-30% of participants will be designated as Winners. All winners will receive a medal and a Certificate of Achievement.

A Champion Trophy will be awarded to the medalist with the highest score in each section (Mathematics, Science and combined). Every contestant who attends the national round of BdMSO will receive a certificate of participation.

International Round: Participants representing Bangladesh at the international round are eligible for IMSO's global Participant Certificate as a member of the national team. Depending on the international competition's results, contestants may be awarded Gold, Silver, and Bronze medals, as well as prestigious special awards such as "Best Theory," "Best Exploration," or "Best Experiment."

9. Fee & Registration Process

Prospective participants are required to register online to ensure their participation in the National Round of BdMSO. For Registration please visit: www.bdmsso.org.

Please note participation in the BdMSO process involves the following fees at different stages:

- **National Olympiad:** A registration fee is applicable for the initial online registration to sit for the National Olympiad, ranging 1000-1500 BDT per student.
- **National Residential Camp:** Qualifiers (Top 20%) invited to the residential camp must pay a camp participation fee, ranging 3000-5000 BDT per student.
- **Team Selection Test:** Students qualifying for the Team Selection Test must pay a TST Examination Fee of 1,000 BDT per subject.
- **International Participation & Prep Fee:** Costs associated with Phase B (Team Preparation) and Phase C (International Participation) will be merged into a single comprehensive fee. This fee covers the IMSO Preparatory Camp, Visa processing, Airfare, full international participation costs, and secretariat/administrative expenses. The final amount will be agreed upon and announced following the TST.
- **Financial Aid:** Participants are encouraged to seek independent sponsorship for their fees. However, the organizing committee may consider partial waivers of up to 50% for the National Round, Camp, or TST fees for gifted students demonstrating significant financial need.

10. Organizing Team

The BdMSO is governed by a structured organizational hierarchy to ensure smooth execution across all stages. The management is overseen by the Organizing Bodies (SPSB and BdOSN), followed by the Program Manager or Team Leader.

Throughout the progression from the National Olympiad to the International Round, two dedicated Subject Coordinators (one for Mathematics and one for Science) lead the academic sections. Their responsibilities extend beyond academic duties to serving as Assistant Coordinators for the entire program duration, from the National Round through the International Round, reporting directly to the Program Manager, Team Leader, or overall Olympiad Coordinator. At the operational level, mentors and volunteers support the program management across all academic and logistic functions.

For the international round, the Program Manager/Team Leader and the two Subject Coordinators are typically selected to serve as the official Team Leader and Tutors. In instances where the delegation size is expanded (e.g., 9+9 or 12+12 formations), the organizing team will appoint two Team Leaders and two Tutors for each subject. The organizers also reserve the right to include additional Team Leaders or VIP members within the delegation as representatives of the organizing bodies.

Mentor, Volunteer & Staff Eligibility

To ensure the highest level of integrity, the following eligibility criteria apply to all staff and volunteers:

- **Volunteer Requirements:** General volunteering opportunities are open to students in Grade 11/A-Levels or higher.
- **Academic Mentors/Tutors:** Mentors and tutors for the National Team must possess a proven academic track record or be former Olympiad medalists.
- **Administrative Rights:** The organizers reserve the right to disqualify or reassign any official or volunteer to maintain the neutrality and transparency of the competition.

Conflict of Interest

No individual may serve on the Academic Team (involved in question setting or evaluation) if they have a first-blood relative (child or sibling) participating in any stage of the BdMSO for that year. All team members must sign a conflict-of-interest disclosure.

11. Code of Conduct

To maintain the integrity of the Olympiad, all participants and guardians must adhere to the following:

- **Academic Honesty:** Any form of cheating, plagiarism, or use of unauthorized aids (calculators, smartwatches, mobile phones) during the competition will result in immediate disqualification.
- **Behavior:** Participants are expected to behave respectfully towards judges, invigilators, and fellow competitors. Disruptive behavior may lead to expulsion from the exam hall.
- **Guardian Conduct:** Guardians are not permitted inside the examination halls or restricted camp areas. Interference with the judging process will not be tolerated.
- **Decisions:** All decisions made by the BdMSO organizers regarding scoring, selection, and logistics are final and binding.

12. Media Rights & Data Privacy

By participating in any stage of the BdMSO, participants and their guardians grant the organizers (BdOSN and SPSB) the irrevocable right to capture, record, and use photographs and videos of the participants and attendees. These materials may be reused for promotional purposes, social media, reports, and archive materials. Furthermore, any personal data collected during registration and the competition stages will be used solely for the administration of the Olympiad and related educational initiatives.

13. Disclaimer

The organizers of BdMSO (BdOSN and SPSB) reserve the right to change any rules, regulations, systems, or procedures of the Bangladesh Mathematics and Science Olympiad at any time. All decisions made by the BdMSO organizers shall be considered final and binding.

ANNEX I: BDMSO Syllabus (Mathematics)

1. Arithmetic and Number Theory		
A.	Whole and Rational Numbers	Absolute Value, understanding fractions, decimals and percentages, and converting between them, number base systems.
B.	Properties of Numbers	Even and odd integers, prime and composite numbers, and prime factorization.
C.	Factors and Multiples	Greatest Common Divisor (GCD) and Least Common Multiple (LCM).
D.	Divisibility and Sequences	Divisibility rules, remainders and basic modular arithmetic, cyclic patterns, ordering of numbers, and evaluating complex arithmetic & geometric sequences.
E.	Ratio and Proportion	Solving real-world problems using ratios, rates and direct/inverse proportions.
2. Geometry and Spatial Sense		
A.	Polygons and Circles	Angle relationships, properties of triangles, quadrilaterals, parallelograms, trapeziums, and circles.
B.	Measurements	Calculating angles, area, and perimeter of various 2D shapes.
C.	Coordinate and Transformational Geometry	Basics of the Cartesian plane, plotting points, grid distances, and spatial transformations: symmetry, reflection, rotation, similarity, and proportion.
D.	3D Geometry	Properties of solid figures, cubes, parallelepipeds, calculating surface area and volume.
E.	Spatial Visualization	Understanding and solving problems involving the nets of 3D shapes.
3. Algebra and Pattern Recognition		
A.	Pattern Recognition	Identifying and predicting terms in complex numeric and geometric sequences.
B.	Basic Algebra	Evaluating algebraic expressions, variables as unknowns, and solving simple linear equations and inequalities.
C.	Advanced Expressions	Introduction to quadratic expressions and basic factorization.
4. Data Analysis, Statistics, and Measurement		
A.	Data Representation	Reading and interpreting tables, bar charts, pie charts, line graphs, and scatter plots.
B.	Statistical Averages	Mean, median, and mode.
C.	Advanced Averages	Understanding how changing individual values in a dataset affects the overall average and range.

5. Combinatorics and Probability

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| A. Counting Techniques | Systematic counting, finding the number of possible ordered pairs, and basic permutations and combinations, the Pigeonhole Principle, and the Inclusion-Exclusion principle. |
| B. Probability | Calculating the likelihood of specific outcomes in random events, understanding independent and dependent events. |

6. Recreational Mathematics

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| A. Mathematical Puzzles | Logic grids, parity, magic squares, cryptarithms, working backward, and critical thinking brainteasers. |
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ANNEX II: BDMSO Syllabus (Science)

1. Scientific Method and Experimental Reasoning

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| A. Scientific Thinking | Observation, forming hypotheses, and drawing conclusions based on evidence. |
| B. Measurements | Units of length, mass, time, temperature and volume.
Calculating area and volume from length; Scientific notation.
Using scientific instruments: accuracy, precision, and estimation. |
| C. Experimental Design and Data Interpretation | Organizing data in tables, interpreting experimental diagrams.
Analyzing graph, Extrapolation and Interpolation. |

2. Life Science

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|------------------------------------|--|
| A. The Basics of Life | Characteristics of living vs. non-living things, the cell as the basic unit of life, movement of materials in and out of cells. |
| B. Plant Science | Photosynthesis, respiration, transpiration, seed germination, pollination, tissue systems, and basic hormone regulation. |
| C. Human Anatomy and Organ Systems | Skeleton and muscles; Olfactory, auditory and visual system.
Brain, spinal cord, nerves, and reflex actions.
Eyes, Ears, Nose; Skin and temperature.
Nutrient absorption, digestive system, renal system, kidneys, waste removal, and the endocrine system. |
| D. Health and Metabolism | General health, nutrition, release of energy from food.
Common diseases and their prevention, and hygiene. |
| E. Microbiology and Genetics | Beneficial vs. harmful bacteria, viruses, microbes, antibiotic resistance, basic DNA structure, inheritance, family traits, Introduction to Punnett Squares, and evolution. |
| F. Ecology and Animal Behavior | Habitats, food chains, food webs, ecosystems, population, lifecycle.
Animal behavior, migration, hibernation, fixed action patterns.
Classifying organisms based on food, anatomy, systematics, reproduction and its habitat. |

3. Physical Science

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| A. Chemistry Fundamentals | Atoms and molecules, the mole concept, common elements, simple chemical formulas, carbon as the building block of life, and chemical reactions in everyday life. |
| B. Matter and Fluids | States of matter; physical, chemical & biological transformation.
Motion, speed, density.
Properties of static fluids and gases. |
| C. Forces and Motions | Gravitational, magnetic, elastic, and frictional forces.
Changes in shape due to force. |
| D. Energy and Thermal Properties | Kinetic and potential energy, heat, sound, energy conversion, renewable energy, temperature, thermometers, heat transfer as conduction, convection, and radiation. |
| E. Light and Sound | Properties of light, color, and vision.
The mechanics of sound and vibration. |
| F. Electricity and Magnetism | Basic applications and models. |

4. Earth and Solar System

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| A. The Solar System | The Sun, planets, asteroids, and comets.
Earth's rotation and revolution, phases of the moon, eclipses.
Natural and artificial satellites. |
| B. Earth System | Structure, Surface, and the process of Earth formation.
Water and carbon cycles.
Air composition, wind, weather, climate, seasons, gravitation, tides, and renewable vs non-renewable resources. |
| C. Natural Calamities | Earthquakes, Tsunamis, Cyclone, El Niño and La Niña. |

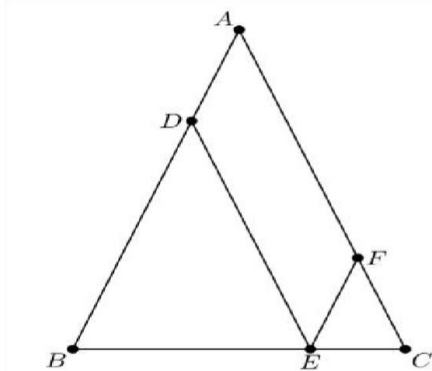
5. Geography and Environmental Science

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|-----------------------------|---|
| A. Global Geography | Continents and oceans, major biomes.
Basic cartography, latitude, and longitude. |
| B. Geography and Bangladesh | Bengal Delta formation, Physiographic regions, river networks.
Haor basin, Sundarbans, monsoons, and coastal salinity. |
| C. Environmental Issues | Greenhouse effect, ozone depletion, climate change, acid rain, Anthropogenic pollution, water crisis, and plastic waste. |
| D. Conservation and Action | Protecting wetlands, forests, and local biodiversity.
Managing natural resources, reforestation vs deforestation, and human-wildlife relations and conflict. |

6. Biotechnology and Modern Science

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|------------------------|--|
| A. Biotechnology | Simple fermentation, basics of genetic engineering and cloning, GMOs, and biofuels. |
| B. Modern Applications | Understanding how emerging scientific and technological developments impact the world and environment. |

ANNEX III: Sample Questions (Mathematics)

Short Answer Problems		
#	Question	Answer
1	In $\triangle ABC$, $AB = AC = 28$ and $BC = 20$. Points D, E, and F are on sides $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$ respectively, such that $\overline{DE}$ and $\overline{EF}$ are parallel to $\overline{AC}$ and $\overline{AB}$ respectively. What is the perimeter of parallelogram? 	
2	How many nonnegative integers can be written in the form $a_7 \cdot 3^7 + a_6 \cdot 3^6 + a_5 \cdot 3^5 + a_4 \cdot 3^4 + a_3 \cdot 3^3 + a_2 \cdot 3^2 + a_1 \cdot 3^1 + a_0 \cdot 3^0$, where $a_i \in \{-1, 0, 1\}$ for $0 \leq i \leq 7$?	
3	Cards are numbered 1,2,...,300. What is the maximum size of a subset S of cards such that no two distinct elements of S sum to 401?	
4	Using digits $\{1,3,5,7,9\}$, find the sum of all 5-digit odd numbers formed?	
5	Find the remainder when the product P is divided by 1000. $P = 1^1 \cdot 2^2 \cdot 3^3 \cdot \dots \cdot 2025^{2025}$	
6	A merchant had some oranges. He sold half of them and 10 more, then half of the rest and 20 more, then half of what was left and 30 more and finally he had 40 oranges left. How many oranges did he have at the beginning?	
7	There are 100 passengers on a train, each paying \$7. They only have 10,15 and 20\$ bills. Trading among themselves they all pay correctly and get the right change. What is the smallest possible number of bills used?	

Essay Problems	
1	A hour digital clock shows times from 1:00 to 12:59 (then repeats). How many times in a 24-hour period is the sum of the digits of the hours equal to the sum of the digits of the minutes?
Ans. 1	
2	You have the digit string 12121 (five digits in that order). You must insert exactly one plus sign (+) and zero or more multiplication signs (x) between the digits to form valid arithmetic expressions produce an odd integer result?
Ans. 2	

Exploration Problems

1.

A set of 13 integers is arranged in nondecreasing order. Let the median be m .

- i. The mean is $m+2$.
- ii. The mode is $2m$, occurring exactly 4 times and uniquely the most frequent value.
- iii. No other integer appears 4 or more times.

Find m and construct one valid distribution.

2.

A palindrome number is a positive integer that can be read the same way in either direction. For instance, 1001, 3443 and 91219 are palindrome numbers. Find all 5-digit palindrome numbers divisible by 44.

3.

Cyclic quadrilateral ABCD has lengths $BC=CD=3$ and $DA=5$ with $\angle CDA=120^\circ$. What is the length of the shorter diagonal of ABCD?

Multiple Choice Questions

1.

Photosynthesis is a chemical process used by plants, algae, and some bacteria to create their own food that gives them the energy to survive. It is the fundamental process that sustains almost all life on Earth. The reaction takes place inside the chloroplasts of a plant's cells, which are tiny organelles that act like little factories.

What is the primary source of all the energy on Earth?

A. Plants B. Soil C. Sun D. Sea

2.

Light is a form of electromagnetic radiation, and while we can see some types, others are completely invisible to us. Light has some properties that determine its characteristics.

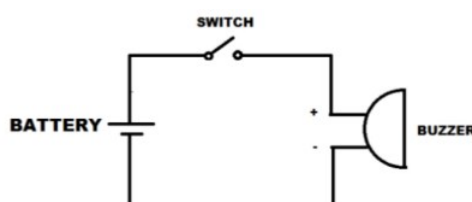
I. Wavelength II. Speed III. Frequency

Which properties of light determine whether we can see it or not?

A. I and II only B. II and III only C. I and III only D. All of the above

3.

The schematic diagram shown below is composed of a battery, a switch, and a buzzer.



In electricity, two concepts are used to describe the flow of current; the conventional current and electron flow. Based on the electron flow notation, which sentence best describes how current flows through a simple circuit when the switch is closed?

**A. Current flows from both the negative and positive sides of the battery to the buzzer.
B. Current flows from the negative side of the battery to the buzzer to the positive side of the battery.
C. Current flows from the positive side of the battery to the negative side of the battery to the buzzer.
D. Current flows from the positive side of the battery to the buzzer to the negative side of the battery.**

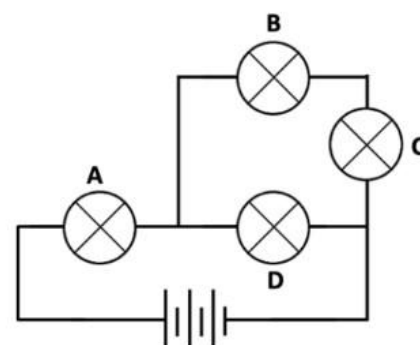
Short Answer Questions

1.
What is the process by which water moves through a plant and evaporates from the aerial parts, mainly through tiny pores called stomata on the leaves?
2.
What is an Eclipse?
3.
List the necessary conditions for heat flow from one object to another.

Brief Answer Questions

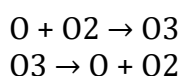
1.
Four identical light bulbs, A, B, C and D are connected as shown on the figure.

A switch S is placed along the circuit such that when S is closed, all the bulbs will light up but when S is opened, only bulbs A and D will light up.

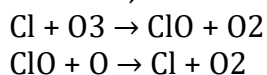


- I. Draw the circuit in your script and locate the possible location of S by drawing a circle on it.
- II. When switch S is closed, identify which light bulb/bulbs will be the- **a.** brightest **b.** least bright

2.
Scientists have discovered a hole in the ozone layer over the Antarctic region. This is largely caused by the release of chlorofluorocarbons (CFCs) in the stratosphere. Ozone (O₃) is constantly produced and depleted in the stratosphere from molecular oxygen (O₂) and atomic oxygen (O).



However, chlorine atom (Cl) from CFCs reacts with the ozone in the following equation:



- I. Explain the importance of the ozone layer to humans.
- II. CFCs are harmful because they introduce chlorine atoms that react with ozone. Even though ozone is constantly produced and depleted, describe and explain how does a single chlorine atom accelerate the depletion of the ozone layer?
- III. Suggest two ways to prevent the further depletion of the ozone layer.